Assignment Code: DA-AG-018

Anomaly Detection & Time Series | **Assignment**

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 100

Question 1: What is Anomaly Detection? Explain its types (point, contextual, and collective anomalies) with examples.

Answer:

Detects unusual data points.

Types:

Point → Single abnormal value
Contextual → Depends on situation
Collective → Group anomaly

Question 2: Compare Isolation Forest, DBSCAN, and Local Outlier Factor in terms of their approach and suitable use cases.

Answer:

Method	Approach	Best Use Case
Isolation Forest	Isolates anomalies using random trees	Large datasets, high dimensional data
DBSCAN	Density-based clustering	Detect clusters and noise
LOF (Local Outlier Factor)	Compares local density	Detect local anomalies

Question 3: What are the key components of a Time Series? Explain each with one example.

Answer:

Trend
Long-term increase or decrease.
Example: Increasing sales over years.
Seasonality
Repeating patterns over time.
Example: Higher ice cream sales in summer.
Cyclic
Long-term fluctuations (not fixed).
Example: Economic cycles.
Irregular (Noise)
Random variation.
Example: Sudden events like strikes

Question 4: Define Stationary in time series. How can you test and transform a non-stationary series into a stationary one?

Answer: A time series is stationary when its statistical properties (mean, variance) remain constant over time.

How to Test:
Augmented Dickey-Fuller (ADF) Test
How to Make Stationary:
Differencing
Log transformation
Removing trend/seasonality

Question 5: Differentiate between AR, MA, ARIMA, SARIMA, and SARIMAX models in terms of structure and application.

Answer:

Model	Description
AR (AutoRegressive)	Uses past values
MA (Moving Average)	Uses past errors
ARIMA	Combines AR + MA + Differencing
SARIMA	ARIMA + Seasonality
SARIMAX	SARIMA + external variables

Dataset:

- NYC Taxi Fare Data
- AirPassengers Dataset

Question 6: Load a time series dataset (e.g., AirPassengers), plot the original series, and decompose it into trend, seasonality, and residual components

(*Include your Python code and output in the code box below.*)

Answer:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
df = pd.read_csv('AirPassengers.csv', parse_dates=['Month'], index_col='Month')
df.plot(title='Original Time Series')
plt.show()
result = seasonal_decompose(df['#Passengers'], model='multiplicative')
result.plot()
plt.show()
```

Question 7: Apply Isolation Forest on a numerical dataset (e.g., NYC Taxi Fare) to detect anomalies. Visualize the anomalies on a 2D scatter plot.

(*Include your Python code and output in the code box below.*)

Answer:

```
from sklearn.ensemble import IsolationForest
import matplotlib.pyplot as plt
import pandas as pd
df = pd.read_csv('nyc_taxi.csv')
X = df[['fare_amount', 'trip_distance']]
model = IsolationForest(contamination=0.05)
df['anomaly'] = model.fit_predict(X)
plt.scatter(df['fare_amount'], df['trip_distance'], c=df['anomaly'])
plt.xlabel('Fare')
plt.ylabel('Distance')
plt.title('Anomaly Detection')
plt.show()
```

Question 8: Train a SARIMA model on the monthly airline passengers dataset. Forecast the next 12 months and visualize the results.

(*Include your Python code and output in the code box below.*)

Answer:

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv('AirPassengers.csv', parse_dates=['Month'],
index_col='Month')
model = SARIMAX(df['#Passengers'], order=(1,1,1), seasonal_order=
(1,1,1,12))
result = model.fit()
forecast = result.forecast(steps=12)
plt.plot(df.index, df['#Passengers'], label='Original')
plt.plot(forecast.index, forecast, label='Forecast', color='red')
plt.legend()
plt.show()
```

Question 9: Apply Local Outlier Factor (LOF) on any numerical dataset to detect anomalies and visualize them using matplotlib.

(*Include your Python code and output in the code box below.*)

Answer:

```
from sklearn.neighbors import LocalOutlierFactor
import matplotlib.pyplot as plt
import pandas as pd
df = pd.read_csv('data.csv')
X = df[['feature1', 'feature2']]
lof = LocalOutlierFactor(n_neighbors=20)
df['anomaly'] = lof.fit_predict(X)
plt.scatter(df['feature1'], df['feature2'], c=df['anomaly'])
plt.title('LOF Anomaly Detection')
plt.show()
```

Question 10: You are working as a data scientist for a power grid monitoring company. Your goal is to forecast energy demand and also detect abnormal spikes or drops in real-time consumption data collected every 15 minutes. The dataset includes features like timestamp, region, weather conditions, and energy usage.

Explain your real-time data science workflow:

- How would you detect anomalies in this streaming data (Isolation Forest / LOF / DBSCAN)?
- Which time series model would you use for short-term forecasting (ARIMA / SARIMA / SARIMAX)?
- How would you validate and monitor the performance over time?
- How would this solution help business decisions or operations?

(*Include your Python code and output in the code box below.*)

Answer:

1. Anomaly Detection
Use:
Isolation Forest (fast)
LOF (local anomalies)
DBSCAN (cluster-based)

2. Forecasting Model
Use:
SARIMA or SARIMAX (handles seasonality + external factors)

3. Validation & Monitoring
Metrics: RMSE, MAE
Continuous monitoring
Retrain model regularly

4. Business Impact
Detect power spikes early
Prevent outages
Improve energy planning
Reduce operational cost